HW8

Naomi Liftman

2025-11-23

```
library(nnet)
library(lmtest)
```

Problem 1

1. Fit the log-linear model of independence and determine if it fits the data well.

```
#create the table
crones = c(2037, 1757, 958, 218)
crones.array = array(crones, c(2,2))
#create independent model
indepmodel = loglin(crones.array, margin=list(1,2),fit=TRUE,param=TRUE)
```

```
## 2 iterations: deviation 0
```

```
indepmodel
```

```
## $lrt
## [1] 312.4785
##
## $pearson
## [1] 289.1536
##
## $df
## [1] 1
##
## $margin
## $margin[[1]]
## [1] 1
##
## $margin[[2]]
## [1] 2
##
##
## $fit
##          [,1]     [,2]
## [1,] 2286.324 708.6761
```

```
## [2,] 1507.676 467.3239
##
## $param
## $param$`(Intercept)`
## [1] 6.940862
##
## $param$`1`
## [1]  0.2081879 -0.2081879
##
## $param$`2`
## [1]  0.585651 -0.585651
```

The model we have fit has $u = 6.940862, u_{1(1)} = 0.2081879, u_{1(2)} = -0.2081879, u_{2(1)} = -0.585651, u_{2(2)} = 0.585651$. To check for the goodness of fit of this model we can see the Chi-Squared value using both G^2 and X^2.

```r
#create the saturated model
satmodel = loglin(crones.array, margin=list(c(1,2)),fit=TRUE,param=TRUE)
```

```
## 2 iterations: deviation 0
```

```r
#testing the p-values
1-pchisq(indepmodel$pearson-satmodel$pearson, indepmodel$df-satmodel$df)
```

```
## [1] 0
```

```r
1-pchisq(indepmodel$lrt-satmodel$lrt, indepmodel$df-satmodel$df)
```

```
## [1] 0
```

What we can see is that both of our chi-squared statistics are equivalent to zero, this means that we reject the null hypothesis and we have evidence to suggest there is a relationship between SNP 2 and Crohns disease.

2. Derive the logistic regression of X1 given X2 from the log-linear model you chose. Provide an interpretation for this regression equation.

```r
#instead of an array, now just make two vectors
disease <- rep(c(0,1), times = c(2037 + 958, 1757 + 218))
genotype <- c(rep(0,2037), rep(1,958), rep(0, 1757), rep(1, 218))

#fit the model
logmod <- glm(disease ~ genotype, family = binomial())
summary(logmod)
```

```
##
## Call:
## glm(formula = disease ~ genotype, family = binomial())
##
## Coefficients:
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.14787    0.03256  -4.542 5.58e-06 ***
## genotype    -1.33248    0.08180 -16.290  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 6679.1  on 4969  degrees of freedom
## Residual deviance: 6366.6  on 4968  degrees of freedom
## AIC: 6370.6
##
## Number of Fisher Scoring iterations: 4
```

Due to the previous logistic model, we know that there is an interaction between the genotype and Crohns disease. What are model tells us is that we have statistically significant evidence to suggest that having genotype Bb or bb decreases the log-odds of having Crohns disease by 1.33248 for our alpha level of .05.

Problem 2

1. Caluclate the asymtotic p-value for testing independence versus interaction.

```
#create the table
crohns_2 = c(2037, 1757, 631, 18, 327, 200)
crohns_2.array = array(crohns_2, c(2,3))
#create saturated model
indep2model = loglin(crohns_2.array, margin=list(1,2),fit=TRUE,param=TRUE)
```

```
## 2 iterations: deviation 4.547474e-13
```

```
#checking the independent p-value
1-pchisq(indep2model$lrt,indep2model$df)
```

```
## [1] 0
```

We get an asymptomatic p-value equivalent to zero, which means that we reject the null hypothesis that these two variables are independent. That means that we do have statistically significant evidence to believe that there is an interaction between SNP 2 and Crohns.

2. Based on your choice of log-linear model, derive the regression of disease given SNP 2.

```
#instead of an array, now just make two vectors
snp <- rep(c(0,1), times = c(2037 + 958, 1757 + 218))
genotype <- c(rep(0,2037), rep(1,958), rep(0, 1757), rep(1, 218))

snp <- rep(c(0,1), times = c(2037 + 958, 1757 + 218))
```

```r
genotype <- c(rep(0,2037), rep(1,631), rep(2, 327), rep(0, 1757), rep(1, 18), rep(2,200))

#fit the model
logmod <- glm(snp ~ genotype, family = quasibinomial())
summary(logmod)
```

```
##
## Call:
## glm(formula = snp ~ genotype, family = quasibinomial())
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.24549    0.03246  -7.562 4.68e-14 ***
## genotype    -0.55330    0.05029 -11.003  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 1.010679)
##
##     Null deviance: 6679.1  on 4969  degrees of freedom
## Residual deviance: 6542.2  on 4968  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

Due to the previous problem, we know there is an interaction between the genotype and Crohns disease. What we can tell from this new model, is that we have statistically significant evidence to suggest that having the genotype Bb decreases the log-odds of you having Crohns disease by 0.55330 and having the genotype bb dicreases your log-odds of having Crohns disease by a further 0.55330. These are all at the alpha level of .05.